BP-R Stage 3: accounting and reference contract

6 October 2026 — reference and upstream-admission A selected; source laws pending

Purpose. Explain the shared conservative accounting and the difference between traffic quantity and disturbance-event metadata. These identities do not adopt a traffic law, prove control leverage or validate gradients. The proposed task has one local controlled ramp and the same open-loop rate permissions in every independently evolving model.

1. Count each part of the population once

Let $\mathcal{C}$ be the disjoint road/ramp cells or lane segments, with $n_c(t)$ their continuous vehicle-equivalent inventories. Let $q_e(t)$ denote waiting upstream of a full modeled entrance e. Physical ramp approach traffic belongs to $\mathcal{C}$; do not also count it as external waiting. If a point queue is used instead of a physical approach, it replaces that approach inventory.

For offered external arrival rate λ_e, actual entrance rate b_e, sink exit rate z_c, and conservative internal transfer rate f_{cd}:

$$\dot{q}_e = \lambda_e - b_e, \tag{1}$$

$$\dot{n}_c = b_c + \sum_d f_{dc} - \sum_d f_{cd} - z_c. \tag{2}$$

Here b_c sums external admissions into cell c; only the actual sink cells have $z_c > 0$. Longitudinal, lateral and merge transfers are all internal f's. Their debits and credits cancel in the population sum. Storage and receiving restrict admissible flows; they do not delete unserved offered demand.

Quantity	Units
$n_c, q_e, M, N^{\text{arr}}, N^{\text{exit}}$	vehicle equivalents
$\lambda_e, b_e, f_{cd}, z_c, r$	vehicle equivalents/second
Density, speed, acceleration	veh/m, m/s, m/s^2
TST	vehicle-equivalent seconds

Define total present population and cumulative arrivals/exits:

$$M(t) = \sum_c n_c(t) + \sum_e q_e(t), \tag{3}$$

$$\dot{M}(t) = \Lambda(t) - Z(t), \qquad \Lambda = \sum_e \lambda_e, \quad Z = \sum_c z_c, \tag{4}$$

$$M(t) = M_0 + N^{\text{arr}}(t) - N^{\text{exit}}(t). \tag{5}$$

The sums involve actual continuous traffic mass, not the number of disturbance descriptors. Counting a descriptor as extra traffic after its mass is already in the fluid would violate (5).

2. Complete system time is the area of present traffic

For a finite offered population, with arrivals ending at T_a, define

$$M_{\text{pop}} = M_0 + \int_0^{T_a} \Lambda(t)\,dt, \qquad \text{TST} = \int_0^{\infty} M(t)\,dt, \qquad J = \text{TST}\,/M_{\text{pop}}. \tag{6}$$

J is mean remaining system time from the declared start, including waiting before admission. Initially present traffic contributes its remaining time; unobserved pre-start time is not reconstructed. Normalization is optional and does not change action rankings when M_{pop} is common.

Why (6)? Each infinitesimal unit of mass contributes one unit of inventory at every instant between its offered arrival and final exit. Integrating these contributions over time counts exactly its system time. Summing first over traffic or first over time gives the same nonnegative area. No additional merge work or recovery penalty is part of this objective: a physical disturbance affects *when traffic exits*, hence the inventory area.

Integrating (5) on a finite horizon and changing integration order gives

$$\text{TST}_T = M_0 T + \int_0^{T} (T - s)\,[\Lambda(s) - Z(s)]\,ds. \tag{7}$$

The weights $T - s$ show why total exits at the endpoint are insufficient. For two actions r_a, r_b with the same initial mass and offered arrivals:

$$\text{TST}_T[r_b] - \text{TST}_T[r_a] = -\int_0^{T} \left(N_b^{\text{exit}}(t) - N_a^{\text{exit}}(t)\right) dt. \tag{8}$$

If the full inventory integrals are finite, take the limit $T \to \infty$. Identical full exit histories imply identical TST even if waiting shifts between the ramp, freeway and entrance. Equal eventual exit totals do not imply equal exit histories or equal TST.

Accounting example, not a freeway witness. Ten units present initially, all exiting at 60 s or at 120 s, give the same eventual exit count but respectively 600 and 1200 vehicle-equivalent seconds. This is an area identity, not evidence of a feasible ramp mechanism or a claimed control benefit.

Clearing tail. After the decision horizon, every model uses the same declared draining policy and remaining forcing. If exact emptying occurs, the integral ends there. Some conservative discretizations empty only asymptotically: then bound the omitted tail instead of requiring a fabricated exact-zero state. For example, a justified bound $M(t) \le B\exp[-\gamma(t - T)]$ for $t \ge T$, with $\gamma > 0$, implies omitted area at most B/γ. This is a conditional error bound, *not* a proposed exponential traffic or recovery law. Finite demand alone does not prove clearance if some traffic can remain blocked.

3. Rate permission and disturbance creation are different maps

A permission $r(t)$ now limits admission to the finite KW ramp approach under approved S3-G2-M.A. It is not the realized ramp entry: propagation and receiving intervene. In a common time-bin accounting convention let q_k be prior waiting mass, a_k the offered mass available in the bin, R_k^a approach receiving in veh/s and Δt its duration. The work-conserving entrance interface admits continuous mass B_k:

$$B_k = \min\{q_k + a_k,\ r_k\Delta t,\ R_k^a\Delta t\}, \qquad q_{k+1} = q_k + a_k - B_k. \tag{9}$$

The approach receives exactly B_k; no mass is rounded. This transcribes the selected cap and availability contract, not a red/green launch law or the source's full numerical solver. Receiving uses the declared current state; the arrival-bin convention must be common to all models. At a point merge, write the joint admissible allocation abstractly as

$$(f_m, f_r) = \mathcal{A}(D_m, D_r, R; X), \qquad 0 \le f_m \le D_m, \quad 0 \le f_r \le D_r, \quad f_m + f_r \le R. \tag{10}$$

X is the declared current physical state; D_m, D_r are demands and R is shared receiving rate. The selected upstream cap applies to entrance release, not automatically to delayed merge entry. Stored ramp traffic can enter above the contemporaneous entrance permission. No allocation formula, priority fraction or capacity reduction is selected by (10).

In the examined hybrid lineage, continuous lane-transfer intensities generate discrete accelerating obstruction events. For one movement, an illustrative deterministic cumulative convention has

$$\eta(t) = \eta_0 + \int_0^t \phi(s)\, ds, \qquad K(t) = \lfloor \eta(t) \rfloor - \lfloor \eta_0 \rfloor. \tag{11}$$

ϕ is actual lane-transfer intensity integrated over the movement's spatial support, in veh/s; η is cumulative vehicle-equivalent transfer progress and K counts unit-threshold disturbance births. At a point merge the relevant transfer is actual merge entry. The phase η_0 must be public if this convention is used. Equation (11) illustrates the working paper's deterministic option; it is *not an adopted ramp actuator or a replacement conservation law.*

A disturbance descriptor can retain target lane, birth time, position, speed and bounded-acceleration history. Its coupling constrains the surrounding flow. Creating one descriptor is not an instruction to replace (2) with integer transfers. The source's exact mass ownership and coupling to moving constraints still require qualification. A fractional remainder cannot simply disappear, and fluid mass cannot be counted twice.

Sensitivity implication only. Event times can move when rate changes move cumulative crossings, even while event count stays fixed. At crossings and changes of constraint topology, an ordinary smooth derivative is not assured. Differentiating sampled floor values alone may miss event-time effects. This flags Stage 4 work; no derivative formula, gradient estimator, smoothing or usefulness result is adopted here.

Sources and transfer status. The continuous conservation and event lead comes from Laval–Daganzo's 2004 working paper, Sections 2–3; the local merge lead is Laval–Cassidy–Daganzo's merge manuscript, Sections 2–4. Gomes–Horowitz 2006, Section 4 motivates the clearance qualification. Equations (1)–(8) are reviewer bookkeeping deductions under the inherited contract; the finite task and reference transfer have not been empirically validated.